

LLM-Assisted Insurance and Survey Automation

Human Review & Quality Assurance Rubric

1. Image & Evidence Accuracy (Score 0-2 per item)

Objective verification that all observations match visible evidence exactly.

- The container number matches the image exactly (no transcription errors).
- Seal numbers are correctly identified and visibly present.
- Unit ID accurately transcribed.
- Displayed temperatures (SETPPOINT, RETURN, SUPPLY) match images.
- Data logger readings match display.
- Cargo branding and labels match visible text.
- Described damage is clearly visible in the image.
- Regulatory or inspection tags accurately described.

2. Quantitative & Technical Accuracy (Score 0-2 per item)

Verification of calculations, conversions, and temperature logic.

- Temperature differentials calculated correctly.
- Fahrenheit-Celsius conversions are accurate.
- Proper frozen cargo standard applied (-18°C / 0°F typical benchmark).
- Technical interpretations (heat load, airflow restriction) are sound.
- No incorrect thermodynamic or food safety conclusions.

3. Evidence Grounding & Traceability (Score 0-2 per category)

Ensures analytical statements are tied to visible evidence or compliance documentation.

- All claims supported by image evidence or documented readings.
- Compliance references match provided PDF documentation.
- Speculative language clearly labeled (e.g., 'suggests', 'may indicate').
- No unsupported microbiological or spoilage claims.
- No hallucinated regulatory standards or entities.

4. Compliance & Regulatory Interpretation (Score 0-2 per category)

Evaluation of correct regulatory and documentation interpretation.

- Correct interpretation of 'KEEP FROZEN' requirements.
- Accurate interpretation of USDA or inspection tags.
- Data logger programmed ranges are correctly interpreted.
- No overstated regulatory violations without clear evidence.

5. Causation & Risk Language Control (Score 0-2 per category)

Ensures neutral, legally appropriate language is used.

- Observational and neutral tone maintained.
- No assignment of liability.
- No definitive causation claims without full evidence.
- Appropriate use of qualifying language (e.g., 'may indicate').

6. Structural Completeness (Score 0-2 per category)

Ensures all required report sections are properly completed.

- Container condition section complete.
- Reefer performance section complete.
- Cargo condition section complete.
- Temperature monitoring section complete.
- Mitigation recommendations provided.
- Extent of loss section appropriately qualified.

7. Professional Report Quality (Score 0-2 per category)

Overall clarity, consistency, and professional presentation.

- Clear and readable language.
- Consistent terminology and measurement units.
- No contradictions between sections.
- No fabricated manufacturers, serial numbers, or standards.

8. Semantic Comparison Evaluation (LLM-as-Judge Framework) (Score 0-2 per category)

- A validated human-written or approved reference report is used as ground truth.
- The AI-generated report is evaluated against the reference.
- The judge LLM compares both reports without being informed which is human vs AI (bias control).
- Evaluation focuses on semantic equivalence, not wording similarity.

Scoring Summary	Category Maximum Points
Image & Evidence Accuracy	16
Quantitative & Technical Accuracy	10
Evidence Grounding & Traceability	10
Compliance & Regulatory Interpretation	8
Causation & Risk Language Control	8
Structural Completeness	12
Professional Report Quality	8
Semantic Comparison Evaluation (LLM-as-Judge Framework)	8
Total Possible	80

Approval Thresholds:

- 65-80: Approved
- 55-64: Minor Revisions Required
- 40-54: Major Revisions Required
- Below 40: Reject / Regenerate Report